claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-aeae4f155c6457604.jsonl`
- SHA-256: `ab98ba1e91246c41dbfbcaf75fa13fec896b8df8f45980048335511dd03b5613`
- Source modified: `2026-05-31T04:18:44+00:00`
- Imported at: `2026-07-05T16:48:21+00:00`
- project: `wf_076beba8-59c`
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`

Transcript

1. user (2026-05-31T04:17:40.133Z)

Source Extractor

```
Research question: "Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision /
document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud
API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date,
narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST
personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-
column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both
born-digital (text layer present) and the harder cases where pdfplumber's
extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies
between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):
1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default.
Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud
options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT,
BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-
commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This
is critical ? many document-AI models and their training datasets are non-commercial (e.g.
LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several
VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo
license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24
GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license
(state explicitly), latest release / maintenance status as of 2025-2026, VRAM need +
quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL
TABLES / multi-column statements:
- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL
(note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 /
2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community
License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR,
EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND
PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),
MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables):
TATR, PP-Structure, UniTable, etc.
```

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly."

Fetch and extract key claims from this source:
URL: https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE
Title: datalab-to/surya ? MODEL_LICENSE (weights) + LICENSE (code)
Found via: OCR + table-structure recognition pipelines (practitioner/implementation) search

Task
1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-05-31T04:17:42.917Z)

I'll fetch the source URL and extract key claims relevant to the research question about local document-AI models for bank statement parsing.

First, let me load the WebFetch tool schema.

3. assistant (2026-05-31T04:17:46.334Z)

I have the WebFetch tool. Now let me fetch the Surya MODEL_LICENSE page.

4. user (2026-05-31T04:17:53.525Z)

Surya Model License Analysis

License Type
The model weights are governed by the **AI PUBS OPEN RAIL-M LICENSE (MODIFIED)**, Version 0.1, dated March 2, 2023.

Commercial Use Restrictions
Yes, the license explicitly restricts commercial use based on revenue and funding thresholds:

Revenue threshold: "for any purpose if You (your employer, or the entity you are affiliated with) generated more than two million US Dollars ($5,000,000) in gross revenue in the prior year"

Funding threshold: "for any purpose if You (your employer, or the entity you are affiliated with) has raised more than two million US dollars ($5,000,000) in total equity or debt funding"

Competitive restriction: The model cannot be used "if You (your employer, or the entity you are affiliated with) provides or otherwise makes available any product or service that competes with any product or service offered by Licensor or its affiliates."

Exemptions apply for personal use or research purposes.

Code License
The document distinguishes between the Model and "Complementary Material" (source code, scripts, documentation), which it explicitly states: "Complementary Material is not licensed under this License," suggesting separate licensing may apply to code.

Key License Names Referenced
- **OpenRAIL-M** (Open Responsible AI License for Models)
- Reference URL: http://licenses.ai/
- Commercial inquiries directed to: https://www.datalab.to/

5. assistant (2026-05-31T04:17:56.571Z)

The WebFetch returned some internally inconsistent dollar amounts (it said both "two million US Dollars ($5,000,000)" in the same clause), which suggests the fast model may have garbled the exact figures. Let me verify the precise threshold values and confirm key details by fetching the raw license file directly.

6. user (2026-05-31T04:18:02.078Z)

AI PUBS OPEN RAIL-M LICENSE ANALYSIS

License Name and Version:
"AI PUBS OPEN RAIL-M LICENSE (MODIFIED)" - Version 0.1, March 2, 2023

Revenue Threshold:
"for any purpose if You (your employer, or the entity you are affiliated with) generated more than two million US Dollars ($5,000,000) in gross revenue in the prior year"

Funding Threshold:
"for any purpose if You (your employer, or the entity you are affiliated with) has raised more than two million US dollars ($5,000,000) in total equity or debt funding"

Complementary Material Definition:
"Complementary Material" means the applicable source code and scripts used to define, run, load, benchmark or evaluate the Model, and used to prepare data for training or evaluation, if any."

Competitor Clause:
"for any purpose if You (your employer, or the entity you are affiliated with) provides or otherwise makes available any product or service that competes with any product or service offered by or made available by Licensor or any of its affiliates."

7. assistant (2026-05-31T04:18:06.663Z)

The dollar figure is still ambiguous in both fetches ? the text "two million US Dollars

($5,000,000)" is internally contradictory (the words say two million, the numeral says five million). This is a load-bearing detail for the research question. Let me retrieve the raw bytes directly to resolve the exact figure.

8. user (2026-05-31T04:18:08.732Z)

12. Limitation of Liability. In no event and under no legal theory, whether in tort (including negligence), contract, or otherwise, unless required by applicable law (such as deliberate and grossly negligent acts) or agreed to in writing, shall any Contributor be liable to You for damages, including any direct, indirect, special, incidental, or consequential damages of any character arising as a result of this License or out of the use or inability to use the Model (including but not limited to damages for loss of goodwill, work stoppage, computer failure or malfunction, or any and all other commercial damages or losses), even if such Contributor has been advised of the possibility of such damages.
(a) for any purpose if You (your employer, or the entity you are affiliated with) generated more than two million US Dollars ($5,000,000) in gross revenue in the prior year, except where Your Use is limited to personal use or research purposes;
(b) for any purpose if You (your employer, or the entity you are affiliated with) has raised more than two million US dollars ($5,000,000) in total equity or debt funding from any source, except where Your Use is limited to personal use or research purposes; or

9. assistant (2026-05-31T04:18:13.192Z)

Now I have the exact verbatim text from the raw source. The contradiction is genuinely in the upstream license file itself ? the words say "two million US Dollars" but the parenthetical numeral says "($5,000,000)". This is a real discrepancy in the source document, not a fetch artifact. I'll note this explicitly.

Let me also confirm whether there is a separate code LICENSE and what it is, since the task title references both files.

10. user (2026-05-31T04:18:15.298Z)

Apache License
Version 2.0, January 2004
http://www.apache.org/licenses/

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"Legal Entity" shall mean the union of the acting entity and all other entities that control, are controlled by, or are under common control with that entity. For the purposes of this definition, "control" means (i) the power, direct or indirect, to cause the direction or management of such entity, whether by contract or otherwise, or (ii) ownership of fifty percent (50%) or more of the outstanding shares, or (iii) beneficial ownership of such entity.

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"Object" form shall mean any form resulting from mechanical transformation or translation of a Source form, including but not limited to compiled object code, generated documentation, and conversions to other media types.

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"Contributor" shall mean Licensor and any individual or Legal Entity on behalf of whom a Contribution has been received by Licensor and subsequently incorporated within the Work.

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----HEADER----
AI PUBS OPEN RAIL-M LICENSE (MODIFIED)

Version 0.1, March 2, 2023 (Modified)
http://licenses.ai/

11. assistant (2026-05-31T04:18:20.915Z)

I now have everything verified from the primary source. The picture is clear and confirmed against raw GitHub bytes:

- **Code (`LICENSE`)**: Apache-2.0 ? permissive, commercially safe.
- **Model weights (`MODEL_LICENSE`)**: AI Pubs Open RAIL-M License (Modified), v0.1 ? NOT permissive. Contains commercial-use restrictions tied to revenue/funding thresholds plus a competitor clause. The dollar figure has an internal discrepancy in the upstream file itself.

This directly confirms the research question's premise that "Surya and Marker use GPL-or-commercial dual licensing" is partially inaccurate for the current repo ? it is now Apache-2.0 code + OpenRAIL-M (restricted) weights, not GPL. This is a central, falsifiable finding.

Let me record my structured output.

12. user (2026-05-31T04:18:34.398Z)

Structured output provided successfully

13. assistant (2026-05-31T04:18:44.765Z)

Done. I fetched and verified the Surya licensing files directly from raw GitHub bytes (resolving an ambiguity the WebFetch summary introduced).

Key findings for the research question (muneem's local, commercial-license-safe statement parsing):

Source quality: primary ? the actual license files in the `datalab-to/surya` repo (`LICENSE` and `MODEL_LICENSE`), publish/version date 2023-03-02.

Bottom line on Surya's license split:
- **Code** (`LICENSE`): Apache-2.0 ? permissive, safe.
- **Weights** (`MODEL_LICENSE`): **AI Pubs Open RAIL-M (Modified) v0.1** ? a *restricted* license, NOT permissive. It bars commercial use above revenue/funding thresholds and adds a competitor-exclusion clause.

Relevance to muneem: This confirms (and corrects) the research-question premise. The premise said "Surya and Marker use GPL-or-commercial dual licensing" ? the *current* repo is actually **Apache-2.0 code + OpenRAIL-M weights**, not GPL. Either way, the **weights fail hard-constraint #3** (permissive weights required). For muneem specifically, the personal-use/research carve-out in clauses (a)/(b) means Avi's single-user personal tool *can* use it now, but the weights are unsafe for any future commercialization, and the competitor clause is a permanent restriction regardless of revenue.

One discrepancy flagged explicitly: the upstream license file is internally inconsistent on the dollar figure ? the prose says "two million US Dollars" while the parenthetical numeral reads "($5,000,000)" in both threshold clauses. This is a defect in the source document itself, not a fetch error (verified against raw bytes). I noted it in the relevant claims.

Verified source files:
- `https://raw.githubusercontent.com/datalab-to/surya/master/MODEL_LICENSE` (weights)
- `https://raw.githubusercontent.com/datalab-to/surya/master/LICENSE` (code, Apache-2.0)